DOCUMENTATION

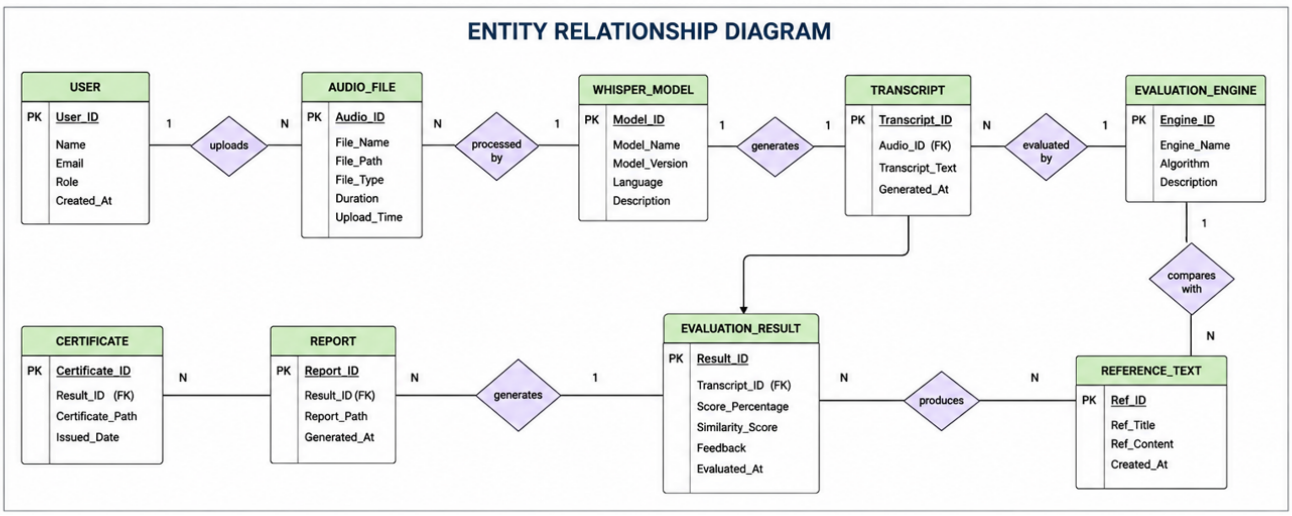
1. Entity Relationship Diagram (ER Diagram)

Uses entities like:

```
User
├── User_ID
├── Name
└── Audio_File

Audio Input
├── Audio_ID
├── File_Name
└── Duration

Transcription
├── Transcript_ID
├── Transcript_Text
└── Timestamp

Evaluation
├── Evaluation_ID
├── Similarity_Score
└── Result

Certificate
├── Certificate_ID
└── Generated_Date
```

Relationship:

User → Uploads → Audio Input
Audio Input → Generates → Transcription
Transcription → Evaluated By → Evaluation
Evaluation → Generates → Certificate



2. Pre-Requisites

Hardware Requirements :

Processor: Intel i3 or above
RAM: 4 GB minimum
Storage: 1 GB free space

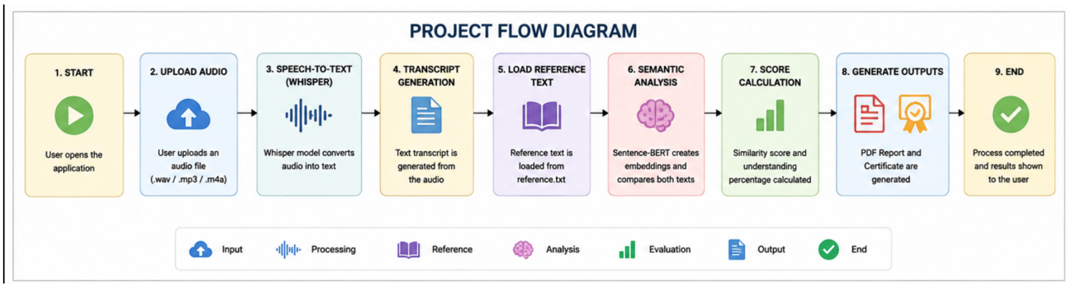
Software Requirements :

Python 3.10+
Streamlit
Whisper
Sentence Transformers
FPDF
Git
VS Code

3. Project Flow

flow chart :

Start
↓
Upload Audio
↓
Speech-to-Text (Whisper)
↓
Extract Transcript
↓
Load Reference Text
↓
Semantic Similarity Analysis
↓
Generate Score
↓
Generate PDF Report
↓
Generate Certificate
↓
End



4. Epic 1: Environment Setup and Dependency Configuration

Development Environment : Windows 10/11
VS Code

Installation :

```
git clone <repository>
cd project
pip install -r requirements.txt
streamlit run app.py
```

Dependencies :

streamlit
openai-whisper
sentence-transformers
torch
fpdf
numpy

5. Epic 2: Core Logic Development

Speech Understanding Module :

The uploaded voice recording is converted into text using OpenAI Whisper.

Evaluation Engine :

The generated transcript is compared with a predefined reference text using Sentence-BERT embeddings.

Similarity Calculation :

Cosine similarity is used to determine concept understanding.

Output :

Similarity Score
Understanding Percentage
Report Generation

6. Epic 3: Streamlit UI Implementation and User Interaction

User Interface Features :

Audio Upload
Transcript Display
Similarity Score Display
PDF Report Download
Certificate Download

User Workflow :

User Uploads Audio
↓
System Processes Audio
↓
Transcript Generated
↓
Concept Evaluation
↓
Results Displayed

7. Epic 4: Testing, Optimization, and Deployment

Testing Types

Functional Testing :

Test Case	Expected Result	Status
Upload Audio	File uploaded	Pass
Speech Conversion	Transcript generated	Pass
Similarity Check	Score displayed	Pass
PDF Generation	Report downloaded	Pass

Optimization : Faster speech processing
Reduced response time
Efficient semantic comparison

Deployment

If hosted: Streamlit Cloud

If local: streamlit run app.py

8. Conclusion

The Voice-Based Concept Understanding Analyser successfully converts speech into text and evaluates conceptual understanding using semantic similarity techniques. The system provides automated assessment, report generation, and certification, making it useful for educational and training environments.

Github Link : https://github.com/ishnu56/Voice-Based-Concept-Understanding-Analyser

Streamlit Link : https://voice-based-concept-understanding-analyser-gckdscqdzdnrwuaohuy.streamlit.app/